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POLICY PROFILE 4.3

RIGHT TO REPAIR AND ANTI-OBSOLESCENCE

A contribution to the Roadmap for Eradicating Poverty Beyond Growth

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EXECUTIVE SUMMARY

Contemporary production models, particularly in electronic products, are structured to maximise corporate profits by intentionally creating "artificial scarcity" through limitations on product durability and repairability. This undermines consumer and workers' rights; it also wastes valuable resources, and imposes "a poverty premium" on low-income households. Manufacturers retain control over repair through proprietary design, "digital lock-in" mechanisms, and restricted access to technical information, making repair difficult and often more expensive than replacement. Our ballooning landfills reveal the state of e-waste that reached nearly 62 million tons in 2022 alone, and is responsible for irreversible health problems, according to the World Health Organisation.

To address these challenges, governments should introduce right to repair and anti-obsolescence policy frameworks that include legally enforceable requirements for product durability and reparability, such as availability of affordable spare parts, independent repair providers, and minimum periods of software support. Planned obsolescence practices, whether through technical, software, or psychological means, should be explicitly prohibited as part of this framework, and public procurement policies should reinforce these standards by prioritising durable and repairable goods.

1. POLICY GOAL

Over the past seventy years, the global economy has been driven by a "growth at all costs" paradigm, leading to rapid digitalisation and a dramatic increase in the throughput of electronic devices. Innovation and falling costs in the domain have indeed broadened access to digital technologies[1]. While innovation and falling costs have broadened access to digital technologies, they have done so through a linear "take-make-waste" model that generates massive environmental and social externalities. This proliferation has produced a largely unregulated stream of electronic waste (e-waste), now recognised as the fastest-growing waste stream in the world. According to the Global E-waste Monitor 2024, the world generated 62 million metric tons of e-waste in 2022, with documented collection and recycling rates growing significantly more slowly, resulting in a widening gap between generation and proper treatment[2].

The core problem lies not merely in the scale of consumption but in the intentional design of "Artificial Scarcity."[3] As described by ecological economists like Jason Hickel, current business models rely on the deliberate limitation of product longevity to ensure repeat sales[4]. This manifests as planned obsolescence, where devices are engineered with "technical obsolescence" (fragile components), "systemic obsolescence" (software incompatibility), and "psychological obsolescence" (marketing-driven dissatisfaction)[5]. These strategies maximise short-term profit margins but undermine consumer rights, breach planetary boundaries, and impose a "poverty premium" on low-income households[6]. This premium is quantifiable. Research in the UK indicates that low-income households pay an additional £490 per year for essential goods because they are forced to purchase cheaper, less durable items that require frequent replacement.[7] It traps vulnerable populations in a cycle of purchasing cheap, short-lived goods that are expensive to operate and impossible to repair.

In line with the "Beyond Growth" approach, this policy seeks to transition the electronics sector from a profit-driven, linear engine of extraction to a circular, fair "Provisioning System." A provisioning system view states that the economy's purpose is to satisfy fundamental human needs, connectivity, communication, and household maintenance, with the minimum necessary resource throughput. By shifting away from a model of relative decoupling (efficiency gains outpaced by volume growth) to one of sufficiency, this policy aims to achieve two primary objectives: first, to extend the physical and functional life of products by prohibiting obsolescence and mandating repairability; and second, to

[1] The Platform for Accelerating the Circular Economy (PACE), *A New Circular Vision for Electronics: Time for a Global Reboot* (World Economic Forum, 2019), 24.
[2] Cornelis P. Baldé et al., *The Global E-Waste Monitor 2024: Electronic Waste Rising Five Times Faster than Documented E-Waste Recycling*, Sustainable Cycles (SCYCLE) Programme (United Nations Institute for Training and Research, 2024), 148, https://ewastemonitor.info/the-global-e-waste-monitor-2024/.
[3] Jason Hickel, 'Degrowth: A Theory of Radical Abundance', *Real-World Economics Review* 87, no. 19 (2019): 54–68.
[4] The Platform for Accelerating the Circular Economy (PACE), *A New Circular Vision for Electronics: Time for a Global Reboot*.
[5] Jeremy Bulow, 'An Economic Theory of Planned Obsolescence', *The Quarterly Journal of Economics* 101, no. 4 (1986): 729–49, https://EconPapers.repec.org/RePEc:oup:qjecon:v:101:y:1986:i:4:p:729-749.; Michael Waldman, 'Durable Goods Theory for Real World Markets', *Journal of Economic Perspectives* 17, no. 1 (2003): 131–54, https://doi.org/10.1257/089533003321164985; Halte à l'Obsolesence Programmée, *Made to Overconsume*, Survey on marketing obsolescence and the influence of commercial strategies (2025), https://www.halteobsolescence.org/publication/marketing-obsolescence-report/.
[6] Halte à l'obsolescence programmée, 'Halte à l'obsolescence Programmée's Contribution to the Development of a Roadmap for Eradicating Poverty Beyond Growth.', Flavie Vonderscher, 28 March 2025.
[7] Dave Beck, 'Poverty Premiums: Cost of Being Poor', in *No Poverty*, ed. Walter Leal Filho et al., Encyclopedia of the UN Sustainable Development Goals (Springer International Publishing, 2020), https://doi.org/10.1007/978-3-319-69625-6_110-1.

close material loops by ensuring end-of-life products are reintegrated into the economy. The environmental logic for this is absolute, particularly for smartphones, where a large share of lifecycle greenhouse gas emissions (often estimated between 70–90%) is attributed to the production phase[8]. At the same time, recycling alone cannot keep pace with growing material demand due to thermodynamic and economic limitations, reinforcing the need for upstream strategies such as product lifetime extension and repair.

The two aspects of this policy profile (anti-obsolescence regulation and allowing the right to repair through various policies) are highly complementary and interdependent. This is why they have to be implemented together. The first will prevent devices from being unrepairable and may force companies to conceptualise products lasting the longest possible. The second will allow consumers to take the lead on owning their goods. The groups most affected by the current system are low-income households, which are disproportionately exposed to abusive practices and bear the financial burden of frequent appliance replacement. By guaranteeing the right to repair, this policy proposal aims to empower consumers, reduce the cost of living and create new opportunities in local repair and refurbishment sectors.

The overall objective is to extend the life of essential goods, reduce electronic waste and establish sustainability as a cornerstone of the European sustainable economy. This policy is desirable because it combines consumer protection with environmental responsibility, contributes to job creation in the repair and reuse sectors, and promotes a culture of sufficiency that underpins long-term resilience and well-being.

2. POLICY FORMULATION

A broad coalition of civil society organisations and UN agencies advocates for this regulation not merely as an environmental measure, but as a critical pathway to reducing material poverty and digital exclusion[9]. This policy profile proposes a holistic transformation of the electronics sector, through a "smart mix" of mandatory regulatory standards ("hard law") and economic incentives designed to retain value within the system. This "policy mix" perspective is critical, as individual measures such as right to repair provisions alone may not be sufficient to significantly extend product lifetimes without complementary instruments addressing product design, business models, consumer practices, and economic incentives. It moves beyond end-of-pipe waste management to address the entire product lifecycle, from design to disposal.

Drawing on Ivan Illich's concept of "Convivial Tools", technology must be intelligible, accessible, and repairable by the user[10]. The policy explicitly prohibits planned obsolescence, building on the 2015 French precedent to make the deliberate shortening of product lifespans a punishable offence. Crucially, it targets the modern mechanism of "digital lock-in" by banning parts pairing (or serialisation). This practice, recently outlawed in Oregon's Senate

[8] The Platform for Accelerating the Circular Economy (PACE), *A New Circular Vision for Electronics: Time for a Global Reboot*.
[9] Diksha Dhiman et al., 'Unlocking Access: The Role of Refurbished Mobile Phones in Digital Inclusion and Sustainable Development', *2024 ITU Kaleidoscope: Innovation and Digital Transformation for a Sustainable World (ITU K)*, 21 October 2024, 1–8, https://doi.org/10.23919/ITUK62727.2024.10772979.
[10] Ivan Illich, *Tools for Conviviality* (Marion Boyars, 2009).

Bill 1596 (signed March 2024), uses software to identify parts through unique identifiers and prevent third-party parts from functioning[11]. Oregon's legislation provides a concrete template for this policy, as it explicitly prohibits manufacturers from using software to "prevent or inhibit" an independent repair provider from installing a replacement part, or from degrading the performance of the device when a non-original part is used.

The core of the proposal lies in "hard law" restrictions that enforce strict design standards upstream. The policy explicitly prohibits planned obsolescence in international law, replicating the 2015 French precedent to make the deliberate shortening of product lifespans a punishable offence. Beyond general prohibitions, the regulation targets specific technical barriers by banning practices such as part-pairing, which blocks independent repair, and the use of non-replaceable parts. It further expands Eco-design regulations to cover mainstream consumer electronics currently overlooked, such as smartphones and laptops, mandating the availability of spare parts and long-term software updates to prevent functional obsolescence[12]. In addition, strengthening legal guarantees and warranty periods should be considered as a complementary measure to incentivise durability. Evidence suggests that aligning mandatory guarantee periods with expected product lifetimes, as defined by technical standards, can be an effective mechanism to promote longer-lasting products and shift incentives for manufacturers.

To correct information asymmetry, the policy establishes mandatory transparency tools that allow consumers to distinguish durable goods from disposable ones. A mandatory Repairability or Durability Index is now displayed at the point of sale, building on successful legislative experiments in France and Belgium.[13] This index transforms repairability into a key competitive characteristic, incentivising manufacturers to improve design scores to appeal to consumers. Alongside this index, manufacturers must be legally obliged to disclose repair manuals and technical diagrams to independent repairers, ensuring that complexity is not used as a pretext to monopolise the aftermarket.

In addition, international trade frameworks and product standards will play a critical role in preventing regulatory arbitrage and ensuring that durability and repairability requirements apply consistently across global value chains. Extending product lifetimes requires coordinated action across multiple system dimensions, including product design, business models, consumer use practices, and reverse logistics systems. Policy interventions must therefore address these interconnected elements rather than focusing on isolated measures.

Finally, the policy reforms the economic architecture of the sector to make it financially viable. The central mechanism for this is the transformation of Extended Producer Responsibility (EPR) schemes from simple waste management fees into active circular

[11] Right to Repair Act, S.B. 1596, Oregon Legislative Assembly 2024 Regular Session, 2024 Oregon Laws (2024), https://legiscan.com/OR/bill/SB1596/2024.
[12] Agata Meysner and Jesus Urios, *The 'Right to Repair': Addressing Social and Environmental Spillovers in the Electrical and Electronic Equipment Sector* (Institute of European Environmental Policy, 2022), 20.
[13] Ronan Groussier et al., *Élaborer un indice de durabilité fiable et ambitieux*, Livre Blanc (Halte à l'Obsolescence Programmée (HOP), 2023); Adèle Chasson et al., *White Paper: Durable and Repairable Products – 20 Steps to a Sustainable Europe*, White Paper (Halte à l'Obsolescence Programmée (HOP), 2020), https://www.halteobsolescence.org/wp-content/uploads/2020/11/White-Paper-HOP-2020.pdf.

economy funds. As outlined by the Right to Repair coalition, this involves the implementation of "eco-modulated" fees, where producers of unrepairable goods pay significantly higher levies, while durable designs are rewarded[14]. Crucially, a minimum share of these EPR revenues must be legally allocated to "Repair Funds," modelled on the French "Bonus Réparation" launched in 2022. These funds provide direct financial incentives that lower the immediate cost of out-of-warranty repairs for consumers, thereby stimulating the local repair economy and internalising the environmental costs of production[15].

3. EXPECTED BENEFITS

If implemented, a right to repair and anti-obsolescence regulation would generate significant social, economic, environmental, and governance benefits. Besides fulfilling the purpose of the report, which is reducing poverty beyond growth, this proposal will address unsustainable consumption patterns and resource-intensive economic dynamics, and their associated environmental and social consequences. This proposal is essential because electronic and electric devices have become essential in our daily lives everywhere in the world, from high-income to low-income countries. Moreover, we often hear in the political discourse that it is essential to combine the environmental and the digital transition. This proposal is an essential aspect of this.

From a social perspective, mandating repairability and affordable access to spare parts empowers consumers and reduces digital insecurity. Vulnerable households would be able to maintain essential electronic devices for longer, lowering the cost of living and narrowing inequalities in access to technology. This is especially relevant in a context of rising cost-of-living pressures. Crucially, in a "beyond growth" framework, poverty reduction is not achieved by pursuing aggregate GDP growth, which often relies on increased consumption. It does it by reducing the cost of satisfying fundamental needs. Current linear models impose a "poverty premium" (or "Vimes' boots" theory), where low-income households are forced to buy cheaper, lower-quality goods that break frequently[16]. It costs them more over time than a durable device would. By mandating durability, this policy dismantles this artificial scarcity. It acts as a pre-distributive measure that protects the disposable income of vulnerable groups, decoupling their well-being from the constant pressure to purchase new goods. Recent biomedical research reinforces the social urgency of this policy; a 2024 World Bank study provided causal evidence that proximity to e-waste dumpsites in West Africa significantly increases neonatal and infant mortality, highlighting that the current linear system is a public health crisis for the Global South[17].

[14] Joan Marc Simon, *Designing EPR to Foster the EU's Competitiveness and Strategic Autonomy*, Ana Oliveira (Zero Waste Europe, 2025).
[15] Right to Repair Europe, *Reforming Extended Producer Responsibility to Promote Repair*, Discussion Paper, with Germanwatch and European Environmental Bureau (European Environmental Bureau, 2024), 20, https://www.umweltbundesamt.de/publikationen.
[16] Terry Pratchett, *Men at Arms: A Discworld Novel* (Corgi, 2013).
[17] Stefania Lovo and Samantha Rawlings, 'The Health Burden of E-Waste: The Impact of E-Waste Dumping Sites on Child Mortality', *The World Bank Economic Review*, 6 December 2024, lhae053, https://doi.org/10.1093/wber/lhae053.

Without coordinated global action, there is a risk that stricter durability and repair standards in high-income countries could shift low-quality products and e-waste flows to lower-income regions, exacerbating environmental and health inequalities.

This proposal can also bring a lot of economic benefits to society. The repair, refurbishment, and recycling sectors are labour-intensive and geographically rooted. However, the economic contribution of repair activities is often underestimated in official statistics, as firms typically report revenues based on their primary activity, which may obscure the true scale of repair-related value creation. By extending product lifespans, the policy creates local employment opportunities and supports small businesses, while reducing dependence on global supply chains for new devices. At the macroeconomic level, it promotes resource efficiency and resilience. Research estimates that circular economy policies could generate approximately 700,000 new jobs in the EU by 2030, as repair activities require significantly more labour per unit of value than highly automated manufacturing[18].

As stated in the introduction, this proposal will clearly have environmental benefits. Extending product use and increasing recycling rates directly reduces premature e-waste generation, lowers raw material demand, and mitigates associated emissions. This contributes to the EU's circular economy and climate neutrality goals by conserving scarce critical raw materials and reducing pressure on global extraction. When talking about governance benefits, it is possible to notice that outlawing planned obsolescence and strengthening consumer rights, the policy enhances trust in markets and ensures more balanced relations between consumers and large technology firms. It aligns with EU policy priorities under the Circular Economy Action Plan, the European Green Deal, and the broader Sustainable Consumption and Production agenda.

4. EVIDENCE, EXAMPLES & BEST PRACTICES

The transition to a repair-based economy is no longer theoretical. Evidence from established laws in France and the US, alongside legislative advances at the EU level, demonstrates that anti-obsolescence regulation is both politically feasible and practically enforceable. Furthermore, empirical research highlights how these policies drive the "beyond growth" objectives of carbon reduction and social equity.

France serves as the primary case study for the feasibility of this regulation. In 2015, it became the first country to explicitly outlaw planned obsolescence, defining it as a criminal offence punishable by fines and potential prison terms. Building on this, French law has required manufacturers to display a repairability index on key electronic devices, a measure that civil society groups identify as a critical step toward durable product design[19]. Empirical evaluation of this index has shown that it effectively influences consumer choice and pushes

[18] Cambridge Econometrics et al., *Impacts of Circular Economy Policies on the Labour Market* (European Commission, 2018), 78, https://circulareconomy.europa.eu/platform/sites/default/files/ec_2018_-_impacts_of_circular_economy_policies_on_the_labour_market.pdf.
[19] Chasson et al., *White Paper: Durable and Repairable Products – 20 Steps to a Sustainable Europe*.

manufacturers to improve product design to achieve higher scores, validating the transparency approach[20].Additional quantitative evidence further supports the effectiveness of such measures. For example, the French "Bonus réparation" scheme supported nearly 1.9 million repair operations over its first three years, demonstrating strong consumer uptake and the scalability of repair incentives.

At the European level, this momentum has been codified into the European Green Deal. The European Commission's Circular Economy Action Plan formally recognised product longevity as a pillar of the EU economy[21]. This has materialised in legislative proposals, including the Ecodesign for Sustainable Products Regulation and the Directive on Common Rules Promoting the Repair of Goods. Policy experts argue that these measures must be complemented by financial incentives, such as redirecting funds from Extended Producer Responsibility (EPR) schemes to directly subsidise repair services[22].

The demand for repair rights extends beyond Europe. In the United States, the state of New York passed the Digital Fair Repair Act in 2022, demonstrating that demanding openness from large technology firms is legally and politically viable in major economies[23]. The desirability of this policy is supported by strong empirical evidence regarding its environmental and social impact. Extending product lifespans is the most effective way to decarbonise the electronics sector. The most recent Global E-waste Monitor warns that e-waste generation is outpacing formal recycling by a factor of five, making waste prevention through repair critical[24]. Furthermore, research by the PBL Netherlands Environmental Assessment Agency highlights that while circular strategies have complex international spillovers, they offer pathways to retain value locally rather than relying on linear imports[25].

Similarly, product-level assessments such as repairability scores (e.g. from iFixit) consistently identify design-related barriers—such as glued components or restricted access—as key obstacles to repair.

Additional empirical insights are provided by datasets from initiatives such as the Open Repair Alliance, which collect data from repair cafés and community repair networks. These data indicate that many electronic devices fail relatively early in their lifecycle and that a significant share of repair attempts are technically feasible. At the same time, they highlight that common barriers to repair are often not technological but structural, including lack of access to spare parts, high repair costs, or product designs that prevent disassembly.

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[20] European Environment Agency, 'Data on Products Evaluated against the French Repairability Index', Circularity Metrics Lab (or You Might Leave This Blank If Zotero Already Captures 'EEA – Circularity Metrics Lab'), 7 February 2024, https://www.eea.europa.eu/en/circularity/thematic-metrics/business/data-on-products-evaluated-against-the-french-repairability-index-1?

[21] European Commission. Directorate General for Communication., *Circular Economy Action Plan: For a Cleaner and More Competitive Europe.* (Publications Office, 2020), https://data.europa.eu/doi/10.2779/05068.

[22] Right to Repair Europe, *Reforming Extended Producer Responsibility to Promote Repair*.

[23] Digital Fair Repair Act, S4104A, New York General Business Law § 399-nn (2022), https://www.nysenate.gov/legislation/bills/2021/S4104.

[24] Baldé et al., *The Global E-Waste Monitor 2024*.

[25] Hester Brink et al., *Potential Effects of Dutch Circular Economy Strategies on Low- and Mid-Income Countries*, no. 4312 (PBL Netherlands Environmental Assessment Agency, 2021), 72.

This suggests that many products are prematurely discarded despite being repairable, reinforcing the importance of policies that address design barriers and improve access to repair services.

5. OBJECTIONS & POTENTIAL BARRIERS & COUNTERARGUMENTS

The implementation of a right-to-repair and anti-obsolescence regulation faces systemic challenges rooted in entrenched economic models, technical complexity, and regulatory fragmentation.

The first problem is the conflict with current linear business models. The most significant barrier is the structural resistance from the technology sector, where current business models rely on "artificial scarcity" to maintain growth rates. As noted by the World Economic Forum, the electronics industry is optimised for volume and rapid replacement cycles, creating a strong financial disincentive for manufacturers to extend product lifespans[26]. Recent game-theoretic economic models, such as those by Yang et al. (2023), warn of a potential "backfire" effect where manufacturers might respond to repair legislation by aggressively lowering the price of new products.[27] This strategy aims to kill the independent repair market by making replacement cheaper than repair, potentially leading to a "lose-lose-lose" outcome of higher environmental waste and lower profits. This reinforces the need for a comprehensive policy mix combining durability standards, minimum software support requirements, and financial incentives for repair.

Incumbents frequently argue that strict durability regulations could stifle innovation or compromise consumer safety. A specific and frequently cited objection is the safety risk associated with high-energy-density components, such as lithium-ion batteries. Manufacturers argue that allowing independent repairers to handle these components increases the risk of "thermal runaway" events and fires, and thus repair should remain restricted to authorized networks[28]. However, civil society groups argue this resistance is often a defence of proprietary ecosystems, where firms use software lock-ins and part-pairing to monopolise the aftermarket and force premature replacement.

The big potential obstacle would be technical and enforcement challenges. Enforcing bans on "planned obsolescence" is legally and technically complex. Effective enforcement will require strengthened market surveillance authorities, harmonised technical standards (including through regional and international standardisation bodies), and the deployment of digital traceability tools such as the Digital Product Passport. The German Environment Agency (UBA) highlights the difficulty in distinguishing between "functional obsolescence" (technical failure) and legitimate cost-saving design choices, thereby increasing the burden

[26] The Platform for Accelerating the Circular Economy (PACE), *A New Circular Vision for Electronics: Time for a Global Reboot*.
[27] Chen Jin et al., 'Right to Repair: Pricing, Welfare, and Environmental Implications', *Management Science* 69, no. 2 (2023): 1017–36, https://doi.org/10.1287/mnsc.2022.4401.
[28] Meysner and Urios, *The 'Right to Repair': Addressing Social and Environmental Spillovers in the Electrical and Electronic Equipment Sector*.

of proof for regulators.[29] Furthermore, the rise of software-defined products introduces new barriers; manufacturers can render perfectly functional hardware obsolete simply by discontinuing software support or security updates. Without harmonised standards across the EU Single Market, national enforcement agencies may lack the capacity to audit complex digital value chains.

Finally, there are obvious economic and cultural barriers. For consumers, the primary objection is the cost of repair relative to replacement. Current market structures often make repair economically unattractive compared to replacement, due to high labour costs, expensive spare parts, and the relatively low price of new products. As a result, right-to-repair provisions alone do not necessarily lead to higher repair rates unless they are complemented by financial incentives and broader structural changes. In addition, regulatory frameworks that restrict repairs to authorised or certified repairers may unintentionally limit accessibility, particularly for low-income consumers, by increasing repair costs and reducing the availability of local repair options. Without financial interventions, such as redirecting Extended Producer Responsibility (EPR) fees to subsidise repair, consumers will remain locked into a "throwaway" culture driven by economic necessity rather than personal preference. [30] However, this transformation requires robust governance mechanisms, transparency, and accountability to ensure that EPR funds are effectively allocated to repair and reuse activities. Additionally, "psychological obsolescence", or "marketing obsolescence" driven by marketing and fast-changing aesthetic trends, remains a powerful cultural barrier that regulation alone cannot fully address. Finally, there is a risk of global fragmentation: without alignment in trade strategy, stricter EU standards could displace low-quality, short-lived goods to the Global South, exacerbating environmental injustice. A potential trade-off that requires careful consideration is that stricter durability requirements and extended warranty obligations may increase upfront product costs, potentially affecting affordability for low-income consumers. However, these higher initial costs are often offset over time by reduced replacement frequency and lower total cost of ownership.

While political and industrial resistance to a right-to-repair regulation is significant, evidence suggests that these challenges are manageable and that the policy offers substantial economic and social dividends. The argument that extending product lifespans stifles innovation is increasingly contradicted by the emergence of circular business models. Rather than causing stagnation, regulatory pressure to eliminate planned obsolescence drives upstream innovation in modularity, durability, and service-based models, such as "Product-as-a-Service", which decouple revenue from resource extraction.[31] Evidence from the French repairability and durability indexes demonstrates that when transparency creates a level

[29] Ines Oehme and Anett Jacob, *Strategies against Obsolescence Strategies against Obsolescence Ensuring a Minimum - Product Ensuring a Minimum - Product - Lifetime and Improving Product - Lifetime and Improving Product - Service Life as Well as - Service Life as Well as -consumer Information* - (German Environment Agency, 2017).
[30] Marco Compagnoni, 'Is Extended Producer Responsibility Living up to Expectations? A Systematic Literature Review Focusing on Electronic Waste', *Journal of Cleaner Production* 367 (September 2022): 133101, https://doi.org/10.1016/j.jclepro.2022.133101.
[31] Simon, *Designing EPR to Foster the EU's Competitiveness and Strategic Autonomy*.

playing field, manufacturers actively innovate to improve their design scores to maintain competitiveness.

Furthermore, the fear of economic loss ignores the employment potential of a repair-based economy. While a shift away from linear production may reduce manufacturing volumes, it generates significant opportunities in the labour-intensive repair, refurbishment, and maintenance sectors. Reviews of the circular economy's labour market impact indicate that these policies can create net positive employment, as repair activities are inherently more labour-intensive and locally rooted than highly automated global manufacturing supply chains.[32] This transition aligns directly with the EU's broader industrial strategy to foster resilience and local job creation.

Concerns regarding the feasibility of enforcement are also mitigated by the increasing sophistication of digital tracking tools. The complexity of monitoring compliance is being addressed through mechanisms such as the Digital Product Passport (DPP), outlined in the European Circular Economy Action Plan, which allows regulators to track component durability and supply chain transparency at scale.[33] Successful national precedents, such as the 2020 French anti-waste law, prove that establishing clear metrics for repairability is not only technically feasible but also popular with the public, thereby strengthening the political legitimacy of the reform against industry lobbying. Regarding safety concerns like thermal runaway, independent experts argue that these risks can be managed through standardised certification for independent repairers rather than through market exclusion.

Finally, political economy challenges can be managed through phased implementation and economic incentives that smooth the industry's transition. Rather than a sudden shock, the policy proposes a "smart mix" of regulation and support, such as using eco-modulated Extended Producer Responsibility (EPR) fees to reward sustainable producers while funding the repair transition.[34] By framing the right to repair as a cost-of-living measure that protects vulnerable households from the "poverty premium" of disposable goods, policymakers can build a broad coalition that transcends traditional environmental vs. industrial divides.

6. IMPLEMENTATION STEPS

The effective rollout of a right-to-repair and anti-obsolescence regulation requires a comprehensive implementation strategy that integrates phased timeframes, coordinated governance levels, and differentiated approaches based on global economic contexts.

The transition toward durable, repairable, and recyclable electronics unfolds across three main time horizons. In the short term, over the next two to three years, the policy should prioritise immediate consumer protection and waste reduction by targeting bans on practices

[32] International Institute for Sustainable Development, *Effects of the Circular Economy on Jobs* (International Institute for Sustainable Development, 2020), 17.
[33] European Commission: Directorate-General for Communication, *Circular Economy Action Plan – For a Cleaner and More Competitive Europe* (Publications Office of the European Union, 2020), https://doi.org/10.2779/05068.
[34] Right to Repair Europe, *Reforming Extended Producer Responsibility to Promote Repair*.

that deliberately shorten product lifespans, including certain forms of planned obsolescence in both hardware and software. Simultaneously, rapid access to spare parts and repair information at fair prices must be mandated. To address the immediate economic barrier, repair bonuses and subsidies should be introduced to lower the cost of repairs, making them a more attractive option than replacement. These regulatory moves should be supported by the launch of EU-wide consumer awareness campaigns highlighting the economic and environmental benefits of repair. In the medium term of around five years, the focus shifts to institutional and fiscal reforms. Mandatory repairability and durability indices, following the French model, must be institutionalised across the EU and displayed at the point of sale. Fiscal policy should be adjusted by reforming VAT regimes to apply reduced rates or exemptions to repair services and second-hand products, while gradually phasing out tax advantages for short-lived products. Additionally, the responsibility of producers should be strengthened through extended producer responsibility (EPR) schemes that require minimum and ambitious targets for repair, reuse, and recycling. In the long term of around ten years, the goal is a structural and cultural transformation. This involves fostering a cultural shift toward durable and sufficient consumption, integrating repair and circular economy skills into vocational training and higher education, and consolidating a European governance framework with an observatory monitoring product lifespans and compliance.

Delivering this phased pathway requires vertical integration across multiple governance levels to ensure consistency and prevent regulatory loopholes. At the local level, municipalities and community organisations serve as the frontline of implementation. They can embed right-to-repair clauses into public procurement rules, establish local repair hubs and repair cafés, and partner with vocational schools to equip technicians for careers in refurbishment. At the national level, ministries, parliaments, and agencies provide the legal and economic backbone by revising consumer codes to strengthen warranties and explicitly outlaw planned obsolescence, reforming tax regimes, and conducting market surveillance to sanction non-compliant manufacturers. At the regional level, trade blocs like the European Union must reinforce harmonised standards, such as the Ecodesign Directive and a unified durability index, and integrate right-to-repair provisions into trade agreements to avoid a "race to the bottom" across markets. At the global level, UN agencies and multilateral frameworks like the OECD must facilitate alignment on e-waste management and create treaties to ensure companies face consistent obligations worldwide, thereby reducing the risk of regulatory arbitrage.

Crucially, this global implementation must adopt a differentiated approach to account for stark variations in economic contexts and prevent the exacerbation of environmental injustice. High-income countries (HICs), which drive the highest per-capita e-waste generation, bear the primary responsibility to reduce premature replacement and internalise e-waste externalities. Their regulatory capacity enables the rapid adoption of repair-first rules, mandatory repairability disclosure, and eco-modulated EPR fees. Producer-side obligations in HICs should also be utilised to fund capacity building in lower-income nations, aligning responsibility with consumption footprints. For middle-income countries (MICs), where informal repair sectors are expanding alongside rising device penetration, policies should focus on formalising and upskilling existing repair ecosystems, tightening import and

treatment standards for second-hand electronics, and securing concessional finance for formal recycling infrastructure. In many of these contexts, repair activities are already widespread but largely informal. Policies should therefore prioritise formalisation, skills recognition, and improved working conditions rather than replacing existing repair ecosystems. In low-income countries (LICs), connectivity increasingly depends on affordable used and refurbished devices, yet these nations bear disproportionate risks from unsafe e-waste handling[35]. The phenomenon of international movements of used electronics creating local health and environmental harms is increasingly described in academic literature as "toxic colonialism."[36] Recent quasi-experimental evidence links proximity to dumpsites to higher neonatal and infant mortality. Therefore, the immediate priority in LICs must be securing safe collection and treatment facilities, establishing basic customs screening to deter illegal e-waste shipments, and providing international support regarding technology transfer to build local monitoring capacity, ensuring the transition to a repair economy is both ecologically sustainable and socially equitable. The implementation pathway can be summarised through a structured roadmap outlining key actions, responsible actors, and indicative timeframes:

Table 1 – Implementation roadmap for right to repair and anti-obsolescence regulation

Action Area	Key Actions	Responsible Level	Timeframe
Policy development	Ban planned obsolescence; introduce right-to-repair legislation; expand ecodesign requirements to consumer electronics	National / Regional	Short-term (2–3 years)
Market incentives	Introduce repair subsidies (e.g. repair bonus); reform EPR schemes with eco-modulation; reduce VAT on repair services and second-hand goods	National	Short–Medium term
Consumer information	Introduce repairability/durability index; ensure access to repair manuals; launch awareness campaigns	National / Local	Short-term
Repair ecosystem	Support local repair hubs and SMEs; formalise informal repair sector; develop vocational training programmes	Local / National	Medium-term

[35] Sarker M. Parvez et al., 'Health Consequences of Exposure to E-Waste: An Updated Systematic Review', *The Lancet Planetary Health* 5, no. 12 (2021): e905–20, https://doi.org/10.1016/S2542-5196(21)00263-1.
[36] Lovo and Rawlings, 'The Health Burden of E-Waste: The Impact of E-Waste Dumping Sites on Child Mortality'.

Action Area	Key Actions	Responsible Level	Timeframe
Industry transition	Ensure availability of spare parts; require long-term software support; promote modular and durable product design	Industry / Regional	Medium–Long term
Global alignment	Harmonise standards; integrate durability requirements into trade frameworks; support capacity building in LICs and MICs	Regional / Global	Long-term

7. INTERNATIONAL HUMAN RIGHTS FRAMEWORK AND INTERNATIONAL COMMITMENTS

The right to repair and anti-obsolescence regulation are closely related with international law, particularly in the face of a prevailing paradigm of electronic device consumption that not only generates greenhouse gas emissions, but ultimately poisons ecosystems and people, thereby contributing to the harmful effects of the triple planetary crisis: the climate emergency, biodiversity loss and pollution.

Within this context, an approach grounded in legal recognition and public policy that directly address this environmental and social problem, would give effect to binding international commitments aimed at safeguarding a range of human rights that are undermined by this unsustainable model of production, including the right to a clean, healthy and sustainable environment,[37] the right to health as recognised in the Universal Declaration of Human Rights (Art. 25) and the International Covenant on Economic, Social and Cultural Rights (Art. 12), as well as the principle of non-discrimination common to both the International Covenant on Civil and Political Rights and the ICESCR (Art. 2).

The production chain of electronic devices may also adversely affect, in an interrelated and interdependent manner, other rights, such as the rights to life, decent work, food, water and adequate housing, as recognised in other treaties. Of particular importance is the right to development, on which the 2030 Agenda for Sustainable Development was grounded and under which SDG 12 seeks to ensure sustainable consumption and production patterns, including through achieving waste reduction by early prevention.[38]

[37] United Nations General Assembly (UNGA), The human right to a clean, healthy and sustainable environment, A/RES/76/300 (2022).
[38] United Nations General Assembly (UNGA), Transforming our world: the 2030 Agenda for Sustainable Development, A/RES/70/1 (2015).

As has been pointed out by the special procedures of the United Nations Human Rights Council, these wastes have differentiated impacts in cases involving, for example, children working in dumpsites and in the dismantling and burning of electronic waste, who, in addition, by living near such sites, are in constant contact with soil and dust usually contaminated with lead and other toxic substances,[39] or women who, in collecting or processing such waste, are exposed to chemicals that have been linked to specific disruptions of the endocrine system and to reproductive health harms.[40] Furthermore, only 20% of the total global volume of this waste is recycled, while the rest is either traded or sent to dump sites, constituting a major source of environmental injustice.[41]

For that reason, international recommendations have called for legislation that provides not only for extended producer responsibility mechanisms, but also for the outright prohibition of planned obsolescence—measures that States must adopt as essential steps to catalyse transformative change in accordance with international human rights law.[42]

At the multilateral level, specific international treaties have also been adopted that crystallise the imperative need to advance a holistic transformation of the sector, as exemplified by the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal, which requires its 191 Parties to reduce waste generation, prevent pollution arising from their management and minimise transboundary movements (Art. 4), while regulating exports and imports and classifying such wastes through its annexes as "hazardous wastes" and "other wastes" (Art. 1). It also establishes the prior informed consent procedure, pursuant to which importing States must authorise such movements before they take place (Art. 6).

In 2019, the so-called Ban Amendment (decisions II/12 and III/1) entered into force under the Convention, legally bringing to an end exports of hazardous wastes intended for final disposal from States members of the Organisation for Economic Co-operation and Development, the European Union and Liechtenstein to States not included in that list. In 2025, several amendments also entered into force (decision BC-15/18), by which hazardous e-waste (for example, waste containing or contaminated with cadmium, lead, mercury, organohalogen compounds and other components), as well as non-hazardous e-waste, were added to the categories of waste covered by the Convention, thereby rendering applicable all the provisions of the treaty and of the Ban Amendment.

Finally, the Review Committee to the Stockholm Convention on Persistent Organic Pollutants also began considering the possible inclusion, in annex C, of polybrominated dibenzo-p-

[39] Special Rapporteur on the implications for human rights of the environmentally sound management and disposal of hazardous substances and wastes, Report on the rights of the child and hazardous substances and wastes, A/HRC/33/41, para. 96, https://docs.un.org/en/A/HRC/33/41.
[40] Report of the Special Rapporteur on the issue of human rights obligations relating to the enjoyment of a safe, clean, healthy and sustainable environment, David R. Boyd, Women, girls and the right to a clean, healthy and sustainable environment, A/HRC/52/33, para. 31, https://docs.un.org/en/A/HRC/52/33.
[41] Special rapporteur on extreme poverty and human rights, Olivier De Schutter, The "just transition" in the economic recovery: eradicating poverty within planetary boundaries, A/75/181, para. 50, https://docs.un.org/en/A/75/181.
[42] Report of the Special Rapporteur on the issue of human rights obligations relating to the enjoyment of a safe, clean, healthy and sustainable environment, David R. Boyd, Women, Business, planetary boundaries, and the right to a clean, healthy and sustainable environment, A/HRC/55/43, para. 74 (b), https://docs.un.org/en/A/HRC/55/43.

dioxins and dibenzofurans and mixed polybrominated/chlorinated dibenzo-p-dioxins and dibenzofurans (PBDD/Fs and PBCDD/Fs), which are often unintentionally generated through activities such as the dismantling and recycling of electronic components (decision POPRC-21/1).[43] This would in turn require the Convention's 185 Parties to take measures to reduce releases of these compounds to a minimum, with a view to their eventual elimination (Art. 5).

All this means that reducing electronic waste is the only adequate pathway to ensuring full compliance with international treaties, both from the standpoint of the technical and capacity-related dimensions incumbent upon the Parties and from that of the protection of persons, having regard to the applicable standards in this field.

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[43] Persistent Organic Pollutants Review Committee of the Stockholm Convention (POPs), Draft risk profile: polybrominated dibenzo-p-dioxins and dibenzofurans and mixed polybrominated/chlorinated dibenzo-p-dioxins and dibenzofurans, UNEP/POPS/POPRC.21/2, paras. 2-5, https://www.pops.int/TheConvention/POPsReviewCommittee/Meetings/POPRC21/Overview/tabid/10273/Default.aspx.

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